

Plot raw EEG waveforms for ASSR recordings.

```
files = {'lab6_exp1_40hz_modulated_10min.mat', ...
        'lab6_trial4_30hz_modulated_10min_2.mat'};

chanNames = {'Fz', 'FC2', 'FC1', 'Cz', 'CP1', 'CP2', 'Pz'};

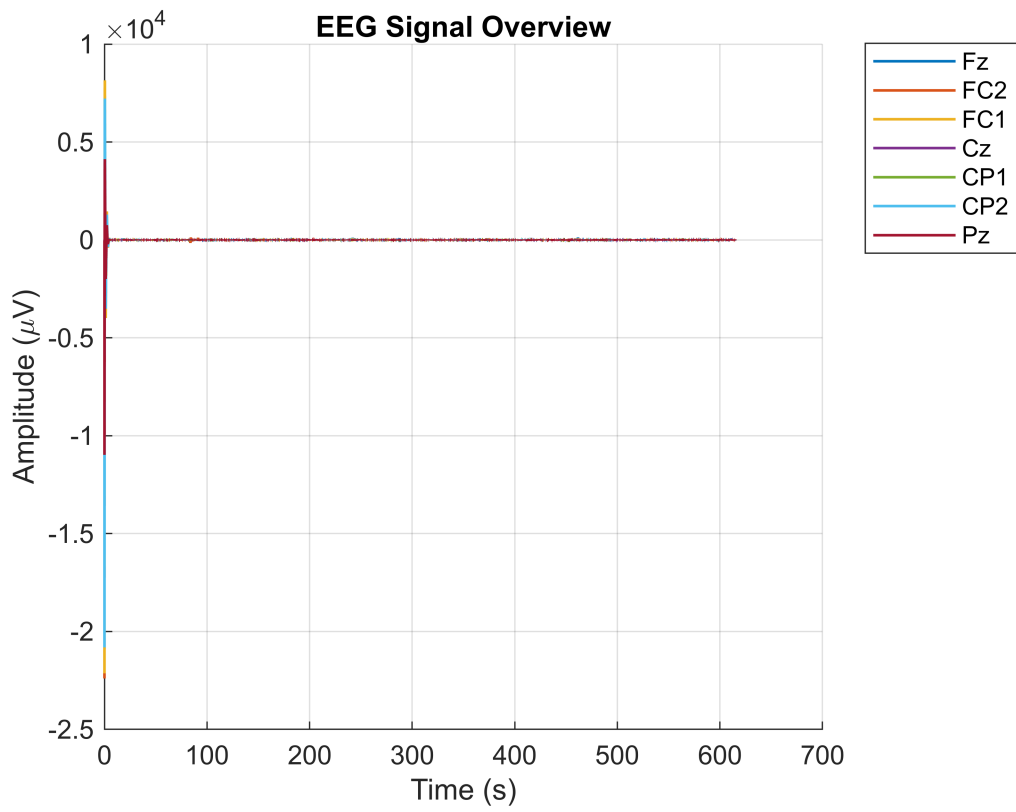
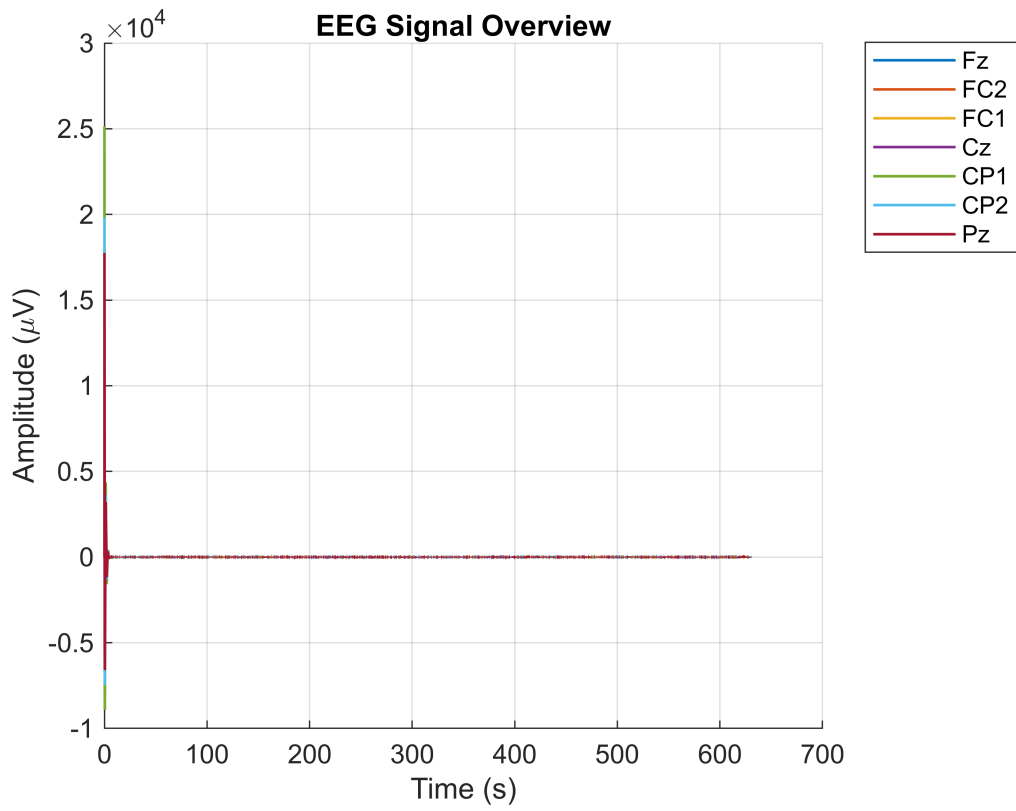
for f = 1:length(files)
    load(files{f});
    y = trigger.Data;
    t = trigger.Time;

    % Cols
    % 1: Fz
    % 2: FC2
    % 3: FC1
    % 4: Cz
    % 5: CP1
    % 6: CP2
    % 7: Pz
    % 8: Trigger (either 0 first/last 15s or 1 throughout modulated tone)
    chanIdx = 1:7;

    EEG = y(:,chanIdx)';
    trg = y(:,8)';

    %get rid of DC offset
    for i = 1:size(EEG,1)
        EEG(i,:) = detrend(EEG(i,:), 'constant');
    end

    %plot
    figure('Color','w');
    hold on;
    for i = 1:length(chanIdx)
        plot(t, EEG(i,:), 'LineWidth', 1);
    end
    hold off; grid on;
    xlabel('Time (s)');
    ylabel('Amplitude (\muV)');
    title('EEG Signal Overview');
    legend(chanNames, 'Location', 'bestoutside');
end
```



The first ~10 seconds of each recording show some pretty visible startup artifacts, which is likely due to some amplifier settling or electrode adjustment. Since in later analyses remove these initial samples by retaining the data when the trigger is equal to 1, this should not affect any results or comparisons across trials.

Plot the time waveform of all the sounds used in your experiments (only 100ms of the signal), and their time-frequency decomposition (a.k.a. a spectrogram. Use MATLAB's spectrogram() function). (1 pts)

```
fs = 44100;           %audio
dur = 10;             % tone duration
t = (0:1/fs:dur)';

%carrier tone
fc = 900;
carrier = sin(2*pi*fc*t);

%40hz
fm40 = 40;
mod40 = 0.5*(1 + sin(2*pi*fm40*t));    %psoitive
am40 = mod40 .* carrier;

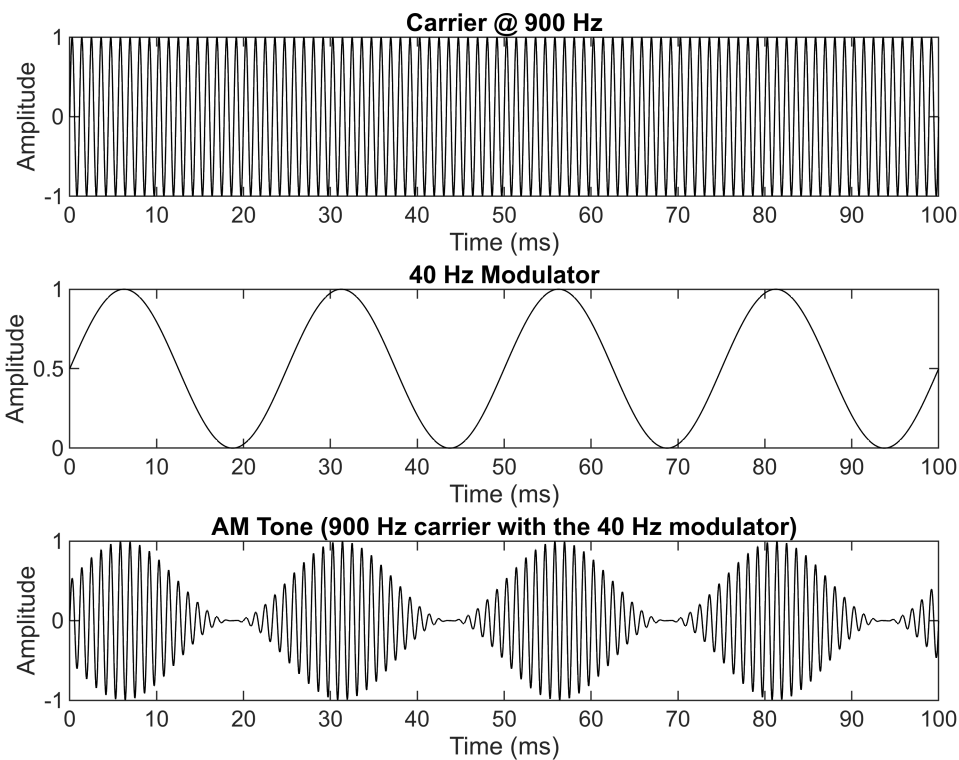
%30hz
fm30 = 30;
mod30 = 0.5*(1 + sin(2*pi*fm30*t));
am30 = mod30 .* carrier;

%100ms
idx100ms = t <= 0.1;

figure('Color','w');
subplot(3,1,1);
plot(t(idx100ms)*1000, carrier(idx100ms), 'k');
xlabel('Time (ms)'); ylabel('Amplitude');
title('Carrier @ 900 Hz');

subplot(3,1,2);
plot(t(idx100ms)*1000, mod40(idx100ms), 'k');
xlabel('Time (ms)'); ylabel('Amplitude');
title('40 Hz Modulator');

subplot(3,1,3);
plot(t(idx100ms)*1000, am40(idx100ms), 'k');
xlabel('Time (ms)'); ylabel('Amplitude');
title('AM Tone (900 Hz carrier with the 40 Hz modulator)');
```



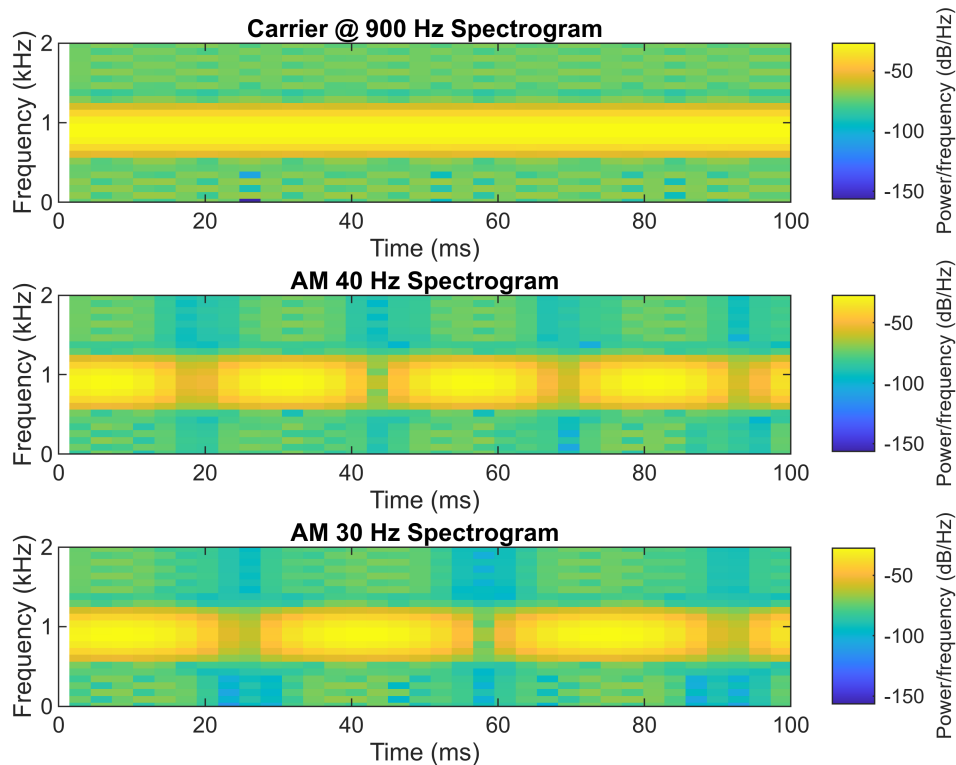
```
%spectrograms
%%over same 100ms and plotting using 0-2kHz y axis
figure('Color','w');

subplot(3,1,1);
spectrogram(carrier,256,128,512,fs,'yaxis');
ylim([0 2]);           % could adjust
xlim([0 0.1]);         %100ms
xticks(0:0.02:0.1);
xticklabels(0:20:100);
xlabel('Time (ms)');
ylabel('Frequency (kHz)');
title('Carrier @ 900 Hz Spectrogram');

subplot(3,1,2);
spectrogram(am40,256,128,512,fs,'yaxis');
ylim([0 2]);
xlim([0 0.1]);
xticks(0:0.02:0.1);
xticklabels(0:20:100);
xlabel('Time (ms)');
ylabel('Frequency (kHz)');
title('AM 40 Hz Spectrogram');

subplot(3,1,3);
```

```
spectrogram(am30,256,128,512,fs,'yaxis');
ylim([0 2]);
xlim([0 0.1]);
xticks(0:0.02:0.1);
xticklabels(0:20:100);
xlabel('Time (ms)');
ylabel('Frequency (kHz)');
title('AM 30 Hz Spectrogram');
```



Load the EEG signal in MATLAB and use the trigger you generated in your s-function to separate the EEG response only when the sound was playing. Since the tone you played was modulated at a particular frequency, the best way to analyze it is in the frequency domain. Take the Fourier transform of each channel, and start your analysis by averaging the magnitude of the FFT over all channels.

Plot the average magnitude. You can expect to see a peak at 40Hz in the EEG data. (1 pt)

```
fs = 352; %the sampling rate used in our experiment (this was the bug that
messed everything up)
files = {'lab6_exp1_40hz_modulated_10min.mat', ...
        'lab6_trial4_30hz_modulated_10min_2.mat'};

for f = 1:numel(files)
```

```

load(files{f}); %
Y = trigger.Data; %N x 8
t = trigger.Time;
EEG = Y(:,1:7)';
trg = Y(:,8)';

%only want when equal to 1
keep = (trg == 1);
if ~any(keep)
    warning('No trigger==1 samples in %s', files{f});
    continue
end
X = EEG(:, keep); % now we have 7 x M

%remove DC per channel
for k = 1:size(X,1)
    X(k,:) = detrend(X(k,:), 'constant');
end

% Okay FFT per channel, the one-sided magnitude, then we take the average
M = size(X,2);
N = 2^nextpow2(M);
FX = zeros(size(X,1), N/2+1);

for k = 1:size(X,1)
    Xk = X(k,:);
    Xk = [Xk, zeros(1, N - M)];
    Fk = fft(Xk, N) / M;
    F1 = Fk(1:N/2+1);
    F1(2:end-1) = 2*F1(2:end-1);
    FX(k,:) = abs(F1);
end

fHz = (0:N/2) * fs / N;
avgMag = mean(FX, 1);

%a little thing to determine the expected modulation frequency from the
%file name, keep resusing
if contains(lower(files{f}), '40hz')
    modFreq = 40;
elseif contains(lower(files{f}), '30hz')
    modFreq = 30;
else
    modFreq = 40; % Just the default
end

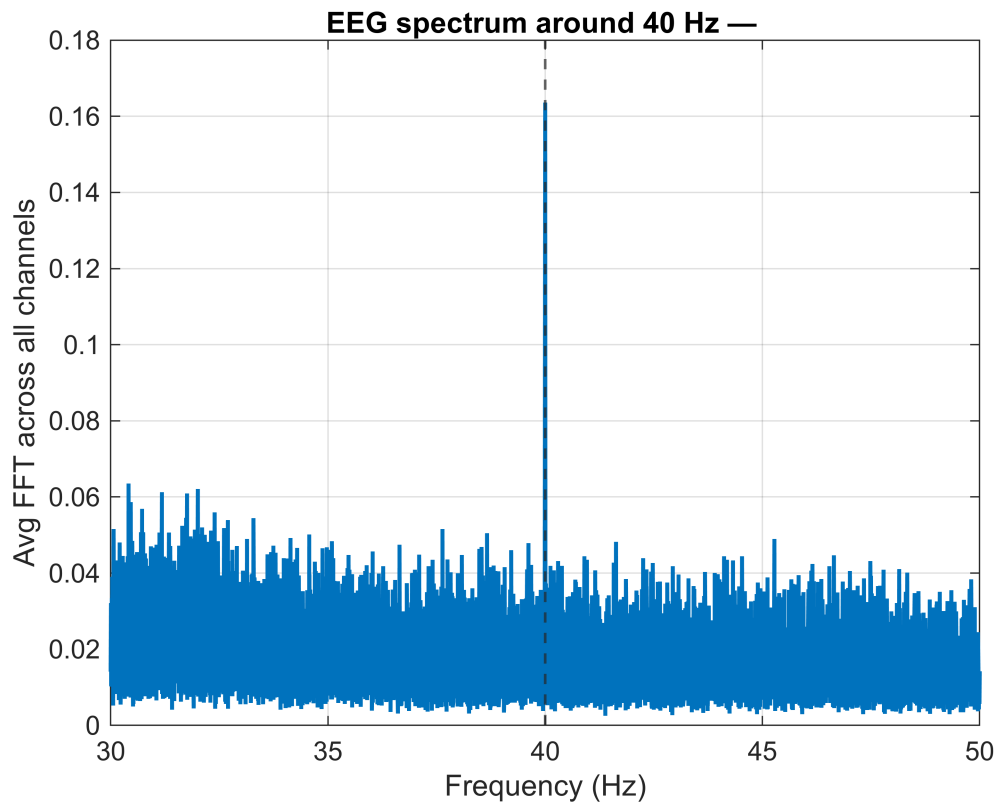
% plot zoomed view ±10 Hz around modulation frequency for a better
% visual
figure('Color','w');
plot(fHz, avgMag, 'LineWidth', 1.5);

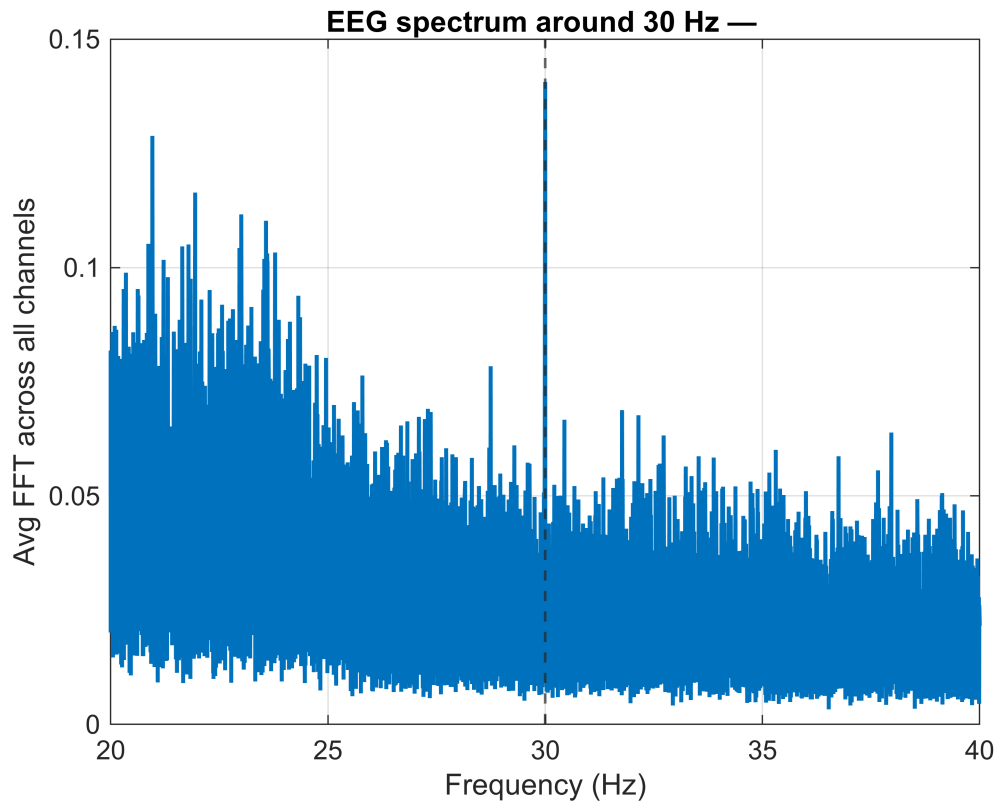
```

```

grid on;
xlim([modFreq-10 modFreq+10]);
xlabel('Frequency (Hz)');
ylabel('Avg FFT across all channels');
title(sprintf('EEG spectrum around %d Hz — %s', modFreq));
xline(modFreq, '--', 'LineWidth', 1);
end

```





If the length of the FFT is L , given the sampling frequency of your EEG signal (256Hz), what sample in FFT corresponds to the modulator frequencies you used (40Hz, 30 Hz)? (1 pt)

```
bin30 = round(30 * N / fs) + 1;
bin40 = round(40 * N / fs) + 1;

fprintf('FFT bin for 30 Hz = %d\n', bin30);
```

FFT bin for 30 Hz = 22343

```
fprintf('FFT bin for 40 Hz = %d\n', bin40);
```

FFT bin for 40 Hz = 29790

```
% lets show the freq. resolution
df = fs / N;
fprintf('Each FFT bin represents %.4f Hz\n', df);
```

Each FFT bin represents 0.0013 Hz

```
%sanity checks
df = fs / N;
f_from_30 = (bin30-1)*df;    %should be 30
f_from_40 = (bin40-1)*df;    %should be 40
```



```
[f_from_30, f_from_40]
```

```
ans = 1×2  
30.0002 39.9999
```

Use the following command to plot the one-sided spectrum: `plot(linspace(0, FS/2,N),f(1:end/2))`, and mark the modulator frequencies on the plots for each condition. Measure the normalized ASSR component by dividing the magnitude of the Fourier transform by its sum. Each Fourier coefficient now represents the relative power at that frequency, instead of overall power. (1 pts)

```
fs = 352;  
files = {'lab6_exp1_40hz_modulated_10min.mat', ...  
        'lab6_trial4_30hz_modulated_10min_2.mat'};  
  
for f = 1:numel(files)  
    load(files{f});  
    Y = trigger.Data;  
    EEG = Y(:,1:7)';  
    trg = Y(:,8)';  
  
    keep = (trg == 1);  
    if ~any(keep)  
        warning('No trigger==1 samples in %s', files{f});  
        continue  
    end  
    X = EEG(:, keep); % [7 x M]  
  
    for k = 1:size(X,1)  
        X(k,:) = detrend(X(k,:), 'constant');  
    end  
  
    % #mostly the same stuff  
    M = size(X,2);  
    N = 2^nextpow2(M);  
    FX = zeros(size(X,1), N/2+1);  
    for k = 1:size(X,1)  
        Xk = [X(k,:), zeros(1, N - M)];  
        Fk = fft(Xk, N) / M;  
        F1 = Fk(1:N/2+1);  
        F1(2:end-1) = 2*F1(2:end-1);  
        FX(k,:) = abs(F1);  
    end  
    fHz = linspace(0, fs/2, N/2+1); % the one-sided frequency axis  
    avgMag = mean(FX, 1); %then averaged across channels  
  
    if contains(lower(files{f}), '40hz'), modFreq = 40;  
    elseif contains(lower(files{f}), '30hz'), modFreq = 30;
```

```

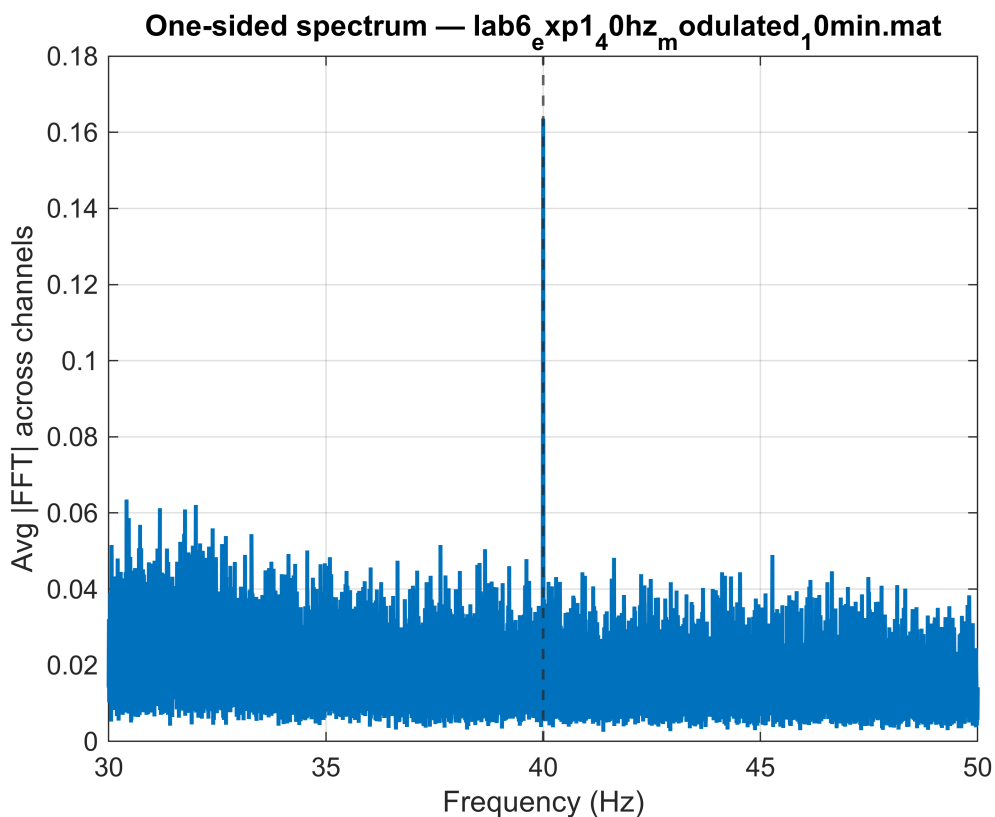
else, modFreq = 40; end

%plot(linspace(0,FS/2,N), f(1:end/2))
figure('Color','w');
plot(fHz, avgMag, 'LineWidth', 1.5); grid on;
xlim([modFreq-10, modFreq+10]);
xlabel('Frequency (Hz)');
ylabel('Avg |FFT| across channels');
title(sprintf('One-sided spectrum - %s', files{f}));
xline(modFreq,'--','LineWidth',1);

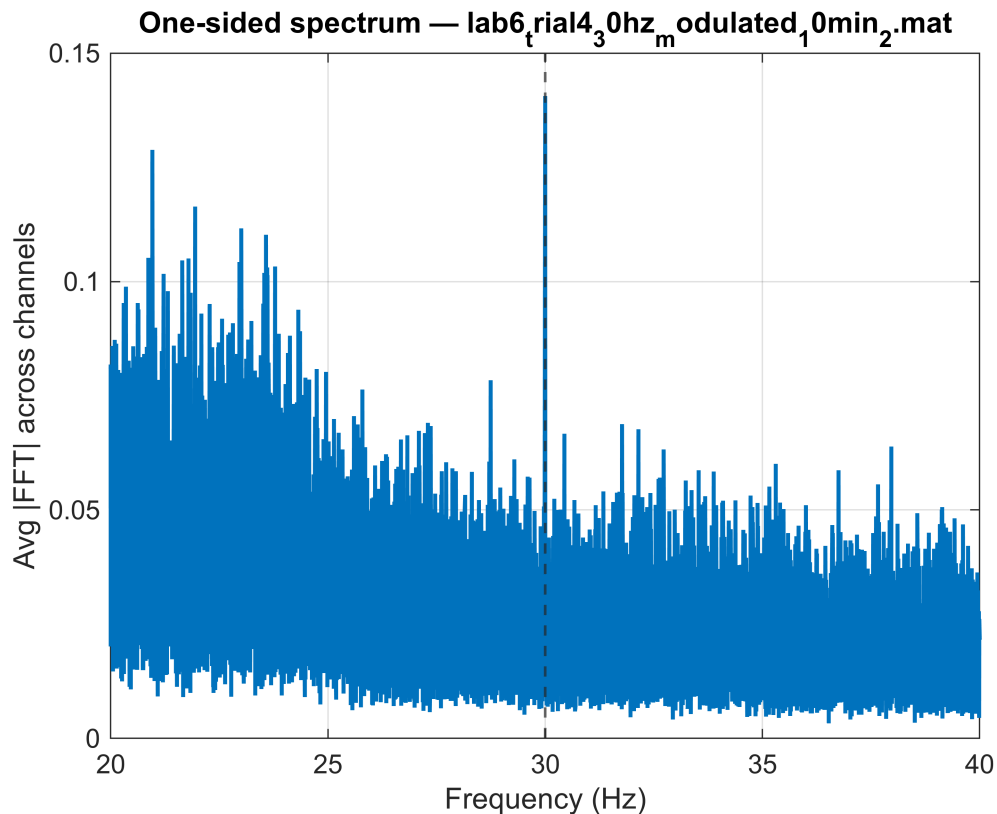
% normalized spectrum and then the normalized ASSR component
normSpec = avgMag / sum(avgMag); % where the each coeff is equal to the
relative power
[~, idx] = min(abs(fHz - modFreq)); % finding the nearest bin to modFreq
normASSR = normSpec(idx);

fprintf('%s: normalized ASSR at %d Hz = %.6f (unitless)\n', ...
        files{f}, modFreq, normASSR);
end

```



lab6_exp1_40hz_modulated_10min.mat: normalized ASSR at 40 Hz = 0.000072 (unitless)



lab6_trial4_30hz_modulated_10min_2.mat: normalized ASSR at 30 Hz = 0.000051 (unitless)

Next, we will determine how much data is needed to get a good ASSR measurement. To answer this important question, repeat your analysis for cumulative 30s intervals (i.e. first 30s, first 60s, first 90s, ...). Display the relative magnitude of the ASSR as a function of signal duration.

```
fs = 352;
interval = 30; % seconds per our step
files = {'lab6_exp1_40hz_modulated_10min.mat', ...
        'lab6_trial4_30hz_modulated_10min_2.mat'};

for f = 1:numel(files)
    load(files{f});
    Y = trigger.Data;
    EEG = Y(:,1:7)';
    trg = Y(:,8)';

    keep = (trg == 1);
    if ~any(keep)
        warning('No trigger==1 samples in %s', files{f});
        continue
    end
    X = EEG(:, keep);
    t = trigger.Time(keep);
```

```

if contains(lower(files{f}), '40hz'), modFreq = 40;
elseif contains(lower(files{f}), '30hz'), modFreq = 30;
else, modFreq = 40; end

% the total available time to use
total_dur = t(end) - t(1);
step_samples = interval * fs;
num_steps = floor(total_dur / interval);

normASSR_vals = zeros(1,num_steps);
durations = (1:num_steps) * interval;

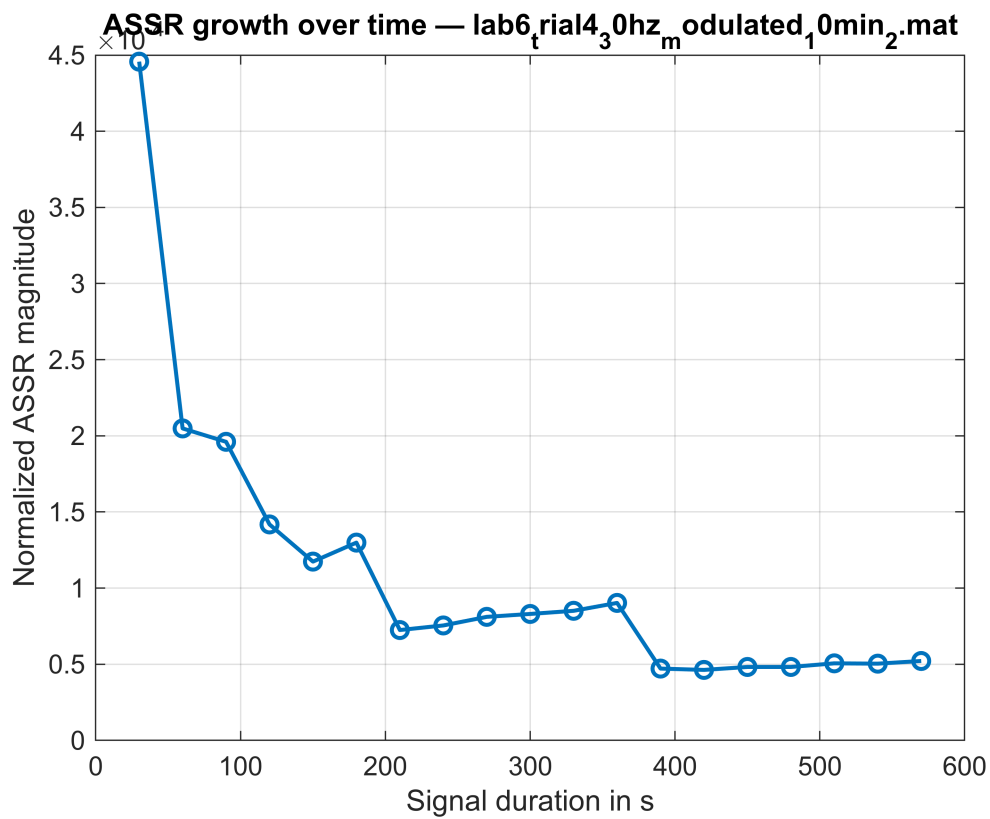
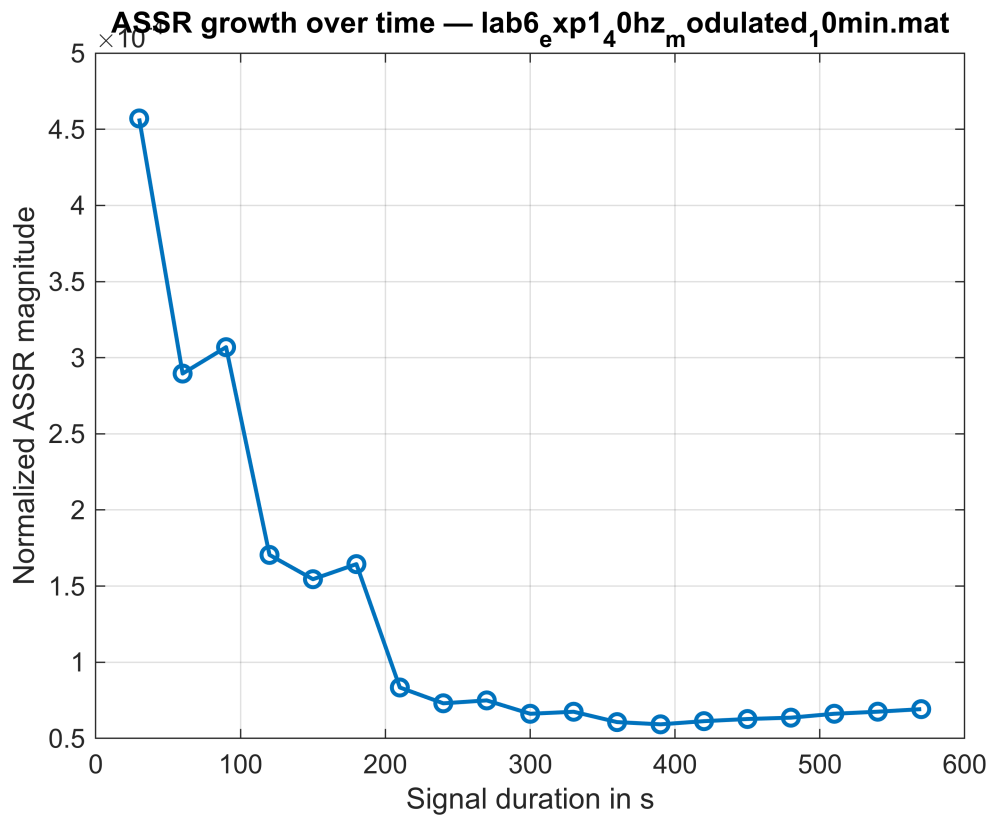
for s = 1:num_steps
    % and then the first s steps
    sample_count = min(size(X,2), round(durations(s) * fs));
    segment = X(:, 1:sample_count);

    % detrend and FFT too
    for k = 1:size(segment,1)
        segment(k,:) = detrend(segment(k,:), 'constant');
    end
    M = size(segment,2);
    N = 2^nextpow2(M);
    FX = zeros(size(segment,1), N/2+1);
    for k = 1:size(segment,1)
        Fk = fft(segment(k,:), N)/M;
        F1 = Fk(1:N/2+1);
        F1(2:end-1) = 2*F1(2:end-1);
        FX(k,:) = abs(F1);
    end
    fHz = linspace(0, fs/2, N/2+1);
    avgMag = mean(FX,1);

    % normalize and then lets extract the ASSR bin
    normSpec = avgMag / sum(avgMag);
    [~, idx] = min(abs(fHz - modFreq));
    normASSR_vals(s) = normSpec(idx);
end

% Then plot the normalized ASSR magnitude v the cumulative. duration
figure('Color','w');
plot(durations, normASSR_vals, 'o-', 'LineWidth', 1.5);
grid on;
xlabel('Signal duration in s');
ylabel('Normalized ASSR magnitude');
title(sprintf('ASSR growth over time - %s', files{f}));
end

```



At what duration can you reliably detect the presence of the ASSR? Does the signal get stronger

the more data you include, or does it reach a plateau after a certain duration? Do you think you collected enough data? If not, how much more do you think you need to reach a plateau? (5 pts)

For the 40 Hz modulation condition, It appears that the normalized ASSR magnitude decreases sharply at first and then happens to stabilize around the 200 second mark. Beyond that point, it seems that the curve remains nearly flat, indicating that around three to four minutes of EEG are sufficient to obtain a reliable 40 Hz steady-state response for these particular conditions.

However, for the 30 Hz condition, we can observe that this trend is more gradual. The response first reaches a temporary plateau between roughly 200 s and 350 s, which is then followed by a lower and more final plateau beginning near our 400s mark. I believe that this two-stage flattening could suggest that the 30 Hz response may require a slightly longer recording perhaps around six to seven minutes (at least as far as our group was recording) before the estimate of the ASSR manages to stabilize.

Overall, I would say that the data tends to show that the ASSR becomes progressively more stable as additional EEG data are averaged. With that said, it seems as if this improvement slows once enough of the cycles of the amplitude-modulated tone have been captured. So, in this lab, which seeks to examine how the auditory steady-state responses manages to emerge from a continuous 30 Hz and 40 Hz amplitude-modulated tones, about 3 minutes of data are enough for 40 Hz and 6–7 minutes for 30 Hz to reach a plateau. Since each of our trials contained roughly 10 minutes of recording, I would say that the amount of data collected was more than enough to achieve a robust and reliable measurement of the ASSR at both of our modulation frequencies.

Measure the ASSR component separately for all channels at the modulator frequency (40Hz in condition 1, 30Hz in condition 2). Use the scalp map to show the relative magnitude of the ASSRs at all electrodes under each experimental condition (30Hz and 40Hz). Which electrodes recorded the largest ASSR component? (1 pt)

```
%it is important that you run addpath('C:\Users\Alexander  
%Speer\Desktop\Columbia Fall 2025\BCI Lab\Lab 6\eeglab14_1_2b') in the CLI before  
you  
%run this code cell.
```

```
%another thing: make sure that you type 'eeglab' in the CLI as well before  
%running this cell. You need both.
```

```
fs = 352;  
files = {'lab6_exp1_40hz_modulated_10min.mat', ...  
        'lab6_trial4_30hz_modulated_10min_2.mat'};  
chanNames = {'Fz', 'FC2', 'FC1', 'Cz', 'CP1', 'CP2', 'Pz'};  
locs_path = 'BCI.locs';  
selLocIdx = [1 2 3 6 7 8 11];  
  
chanlocs = readlocs(locs_path);
```

```
readlocs(): 'loc' format assumed from file extension
```

```

chanlocs = chanlocs(selLocIdx);

nfft = 2^nextpow2(10*fs);
win = round(4*fs);
olap = round(0.5*win);

for f = 1:numel(files)
    S = load(files{f}); y = S.trigger.Data;
    trg = y(:,8) > 0.5;
    X = y(trg,1:7)'; %7 x M
    X = X - mean(X,2);

    % get the welch per channel
    P = zeros(7, nfft/2+1);
    for ch = 1:7
        [pxx, fHz] = pwelch(X(ch,:), hamming(win), olap, nfft, fs, 'onesided');
        P(ch,:) = pxx(:).';
    end

    modFreq = contains(lower(files{f}), '40')*40; if modFreq==0, modFreq=30; end

    %Finding the narrow-band SNR at whatever modFreq were on
    bw_sig = 0.5; % lets ±0.5 Hz the signal band
    bw_nb = [1 3]; % and then noise sidebands 1-3 Hz away
    sig_ix = (fHz>=modFreq-bw_sig) & (fHz<=modFreq+bw_sig);
    nb_ix = ((fHz>=modFreq-bw_nb(2)) & (fHz<=modFreq-bw_nb(1))) | ...
            ((fHz>=modFreq+bw_nb(1)) & (fHz<=modFreq+bw_nb(2)));

    sigP = mean(P(:,sig_ix),2);
    nbP = mean(P(:,nb_ix),2);
    snr_db = 10*log10(sigP./nbP); % 7x1

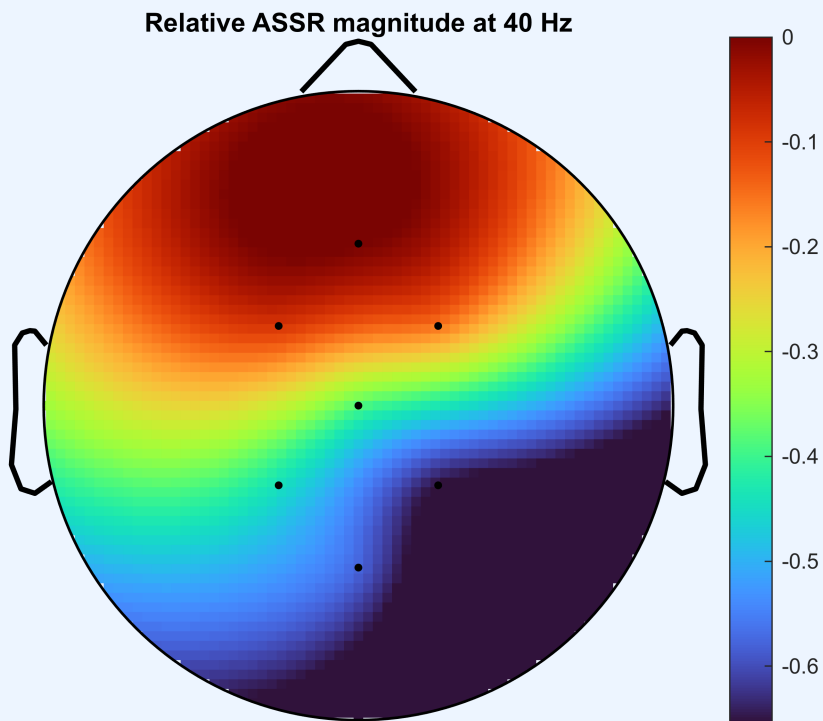
    % have to norm.to the max within this condition for us to actually get a clear
    map
    vals = snr_db - max(snr_db); % we can see that 0 dB is going to be the
    strongest channel

    %get our winners
    [mx, imx] = max(snr_db);
    fprintf('%s: %d Hz ASSR strongest = %s (SNR %.2f dB)\n', ...
            files{f}, modFreq, chanNames{imx}, mx);

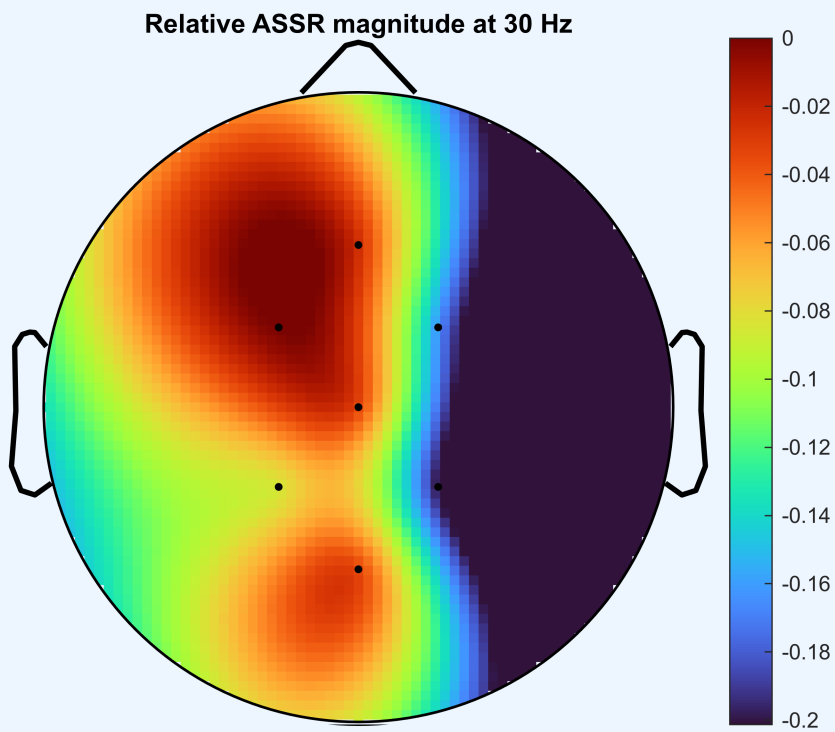
    % IMPORTANT TOPOPLOT
    figure('Color','w','Name',sprintf('ASSR %d Hz SNR (dB rel. max)',modFreq));
    topoplot(vals, chanlocs, 'electrodes','on', 'numcontour', 6, 'style','map',
    'plotrad',0.5);
    caxis([min(vals) 0]); colorbar; title(sprintf('Relative ASSR magnitude at %d
    Hz', modFreq));
end

```

lab6_exp1_40hz_modulated_10min.mat: 40 Hz ASSR strongest = Fz (SNR 1.01 dB)



lab6_trial4_30hz_modulated_10min_2.mat: 30 Hz ASSR strongest = FC1 (SNR -0.05 dB)



At 40 Hz, we can see that the largest ASSR component was recorded at Fz with a SNR $\approx$ 1.0 dB.

At our 30 Hz condition, the largest ASSR component was recorded at FC1 with $\text{SNR} \approx -0.05$ dB.

I believe that the scalp maps show strongest responses over the fronto-central region, which from what I have seen appears to be consistent with auditory steady-state responses generated by the cortical auditory areas.